

How does a neural network learn?

$$x \longrightarrow f_{\theta}(x) \longrightarrow \hat{y} \longrightarrow L \longrightarrow \nabla_{\theta}L \longrightarrow \theta'$$

Supervised learning: data, model and prediction

1. Data: observed input-output pairs

Training dataset

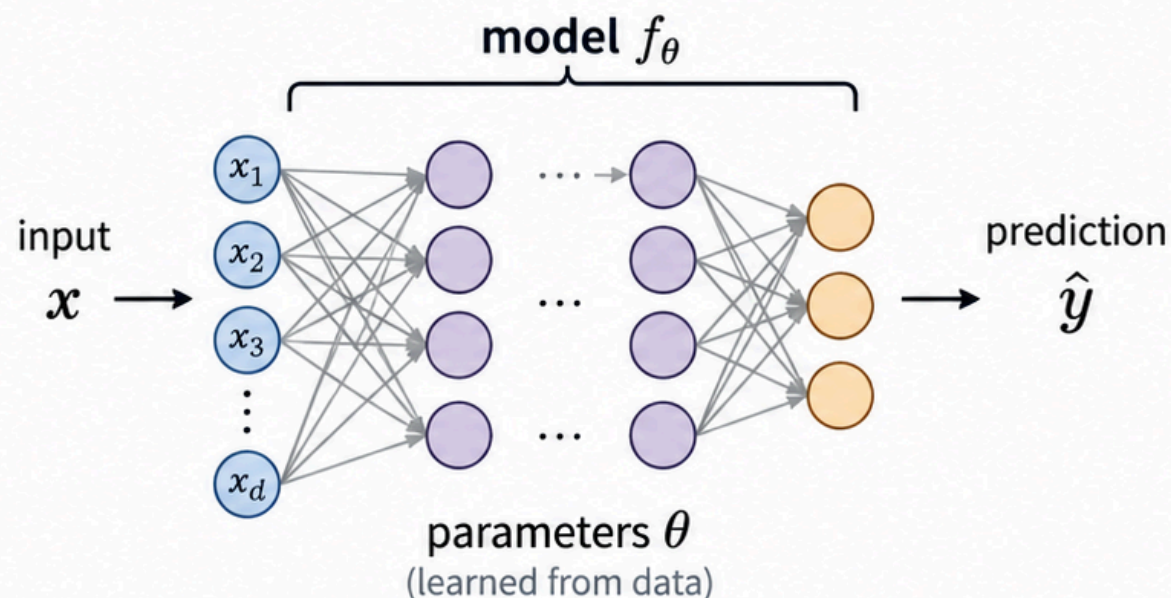
$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$$

x_1	x_2	x_3	...	y
0.5	1.2	3.1	...	0
2.1	0.1	0.3	...	1
1.0	0.8	2.2	...	0
0.3	1.5	1.1	...	1
...	...	...	...	...



2. Model: learn a function that maps x to y

$$\hat{y} = f_{\theta}(x)$$



Examples

- **Classification:** $\hat{y} = f_{\theta}(x) \in \{0, 1, \dots, K - 1\}$
- **Regression:** $\hat{y} = f_{\theta}(x) \in \mathbb{R}$

We observe data

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$$

We start from **observed examples**, not from explicit rules.

From data to a learning problem

1. Real-world problem



We observe inputs x
from the real world
(e.g., images, text, signals, tables).

2. Dataset and splits

Labeled dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$$



(x_1, y_1)
 $y_1 = \text{cat}$



(x_2, y_2)
 $y_2 = \text{dog}$



(x_3, y_3)
 $y_3 = \text{bird}$

$\vdots$

Training set

$\mathcal{D}_{train}$
(e.g., 70–80%)

Validation set

$\mathcal{D}_{val}$
(e.g., 10–15%)

Test set

$\mathcal{D}_{test}$
(e.g., 10–15%)

3. Learning problem

Given

- a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$
- a model $f_{\theta}(x)$
- a loss function $\mathcal{L}(y, \hat{y})$

Learn parameters

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(y_i, f_{\theta}(x_i))$$

Use the learned model to predict

$$\hat{y} = f_{\theta^*}(x)$$

Examples

- **Image classification:** $\hat{y} \in \{0, 1, \dots, K-1\}$
- **Regression:** $\hat{y} \in \mathbb{R}$
- **Multi-label:** $\hat{y} \in \{0, 1\}^K$
- **Sequence prediction:** $\hat{y} = (y_1, \dots, y_T)$

The input

x

The information available **before** making a prediction.

- tabular attributes
- images
- sequences
- signals

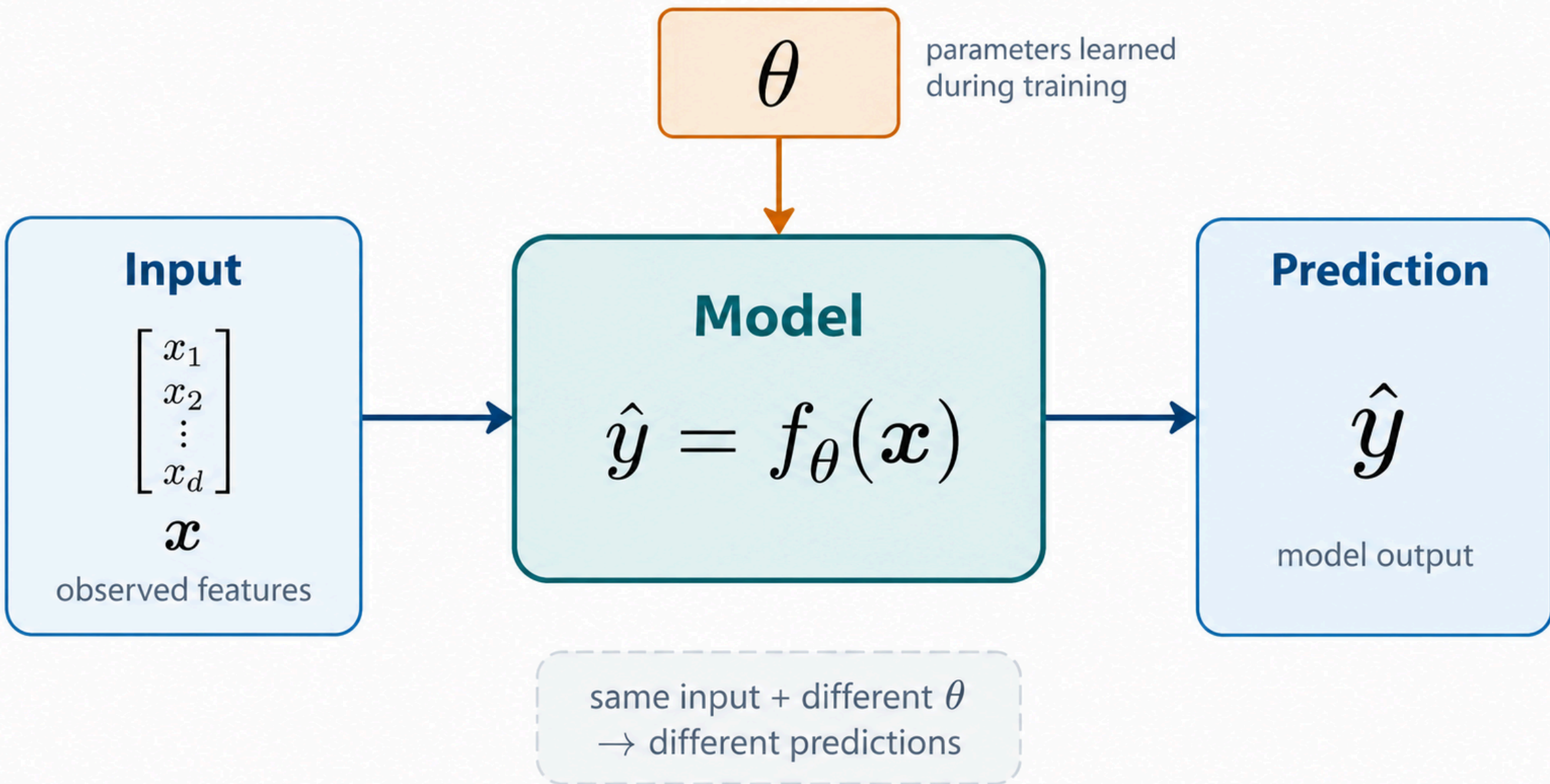
The target

y

The value or class we want to predict.

(x, y)

The target is combined with the prediction to compute the loss during training.



Prediction

$$\hat{y} = f_{\theta}(x)$$

Inference means using the **current parameters** to produce an output.

Is the prediction good?

y

$\hat{y}$

How do we transform the quality of a prediction into something we can optimize?

Loss

$$L(y, \hat{y})$$

For regression, one possibility is:

$$L = (y - \hat{y})^2$$

A loss converts the task objective into a **scalar quantity**.

Loss depends on the parameters

$$\hat{y} = f_{\theta}(x)$$

therefore,

$$L(y, \hat{y}) = L(y, f_{\theta}(x))$$

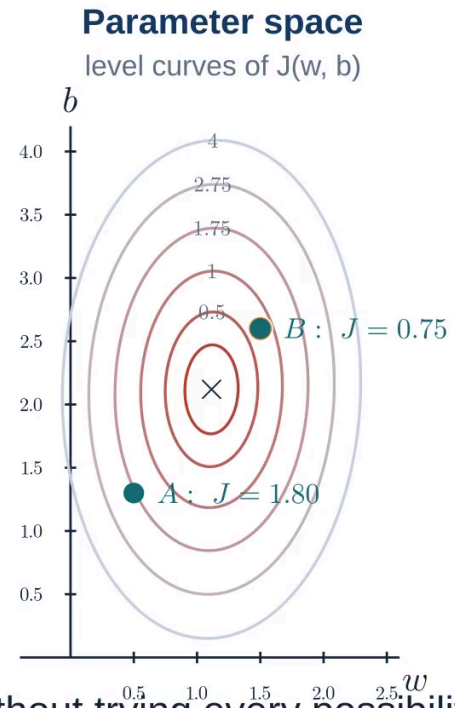
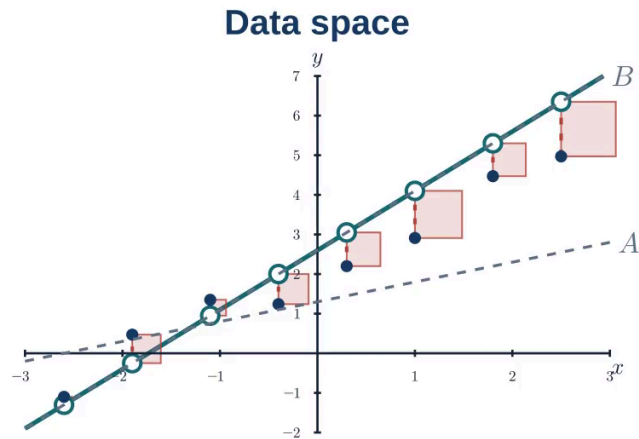
and we can reason about:

$$L(\theta)$$

Parameters, residuals and loss

Parameters, residuals and loss

$$w = 1.50 \quad b = 2.60 \quad \text{MSE} = 0.75$$

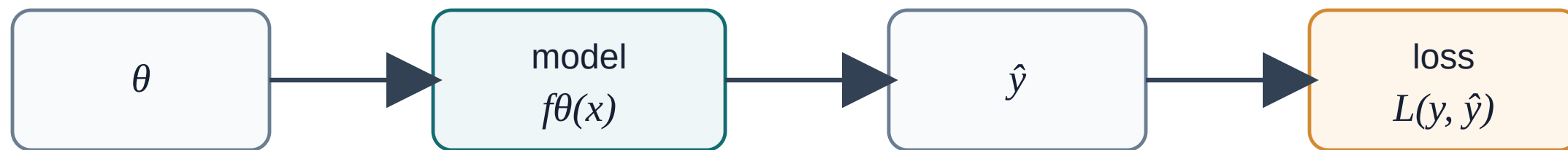


Question: how do we find a better configuration without trying every possibility?

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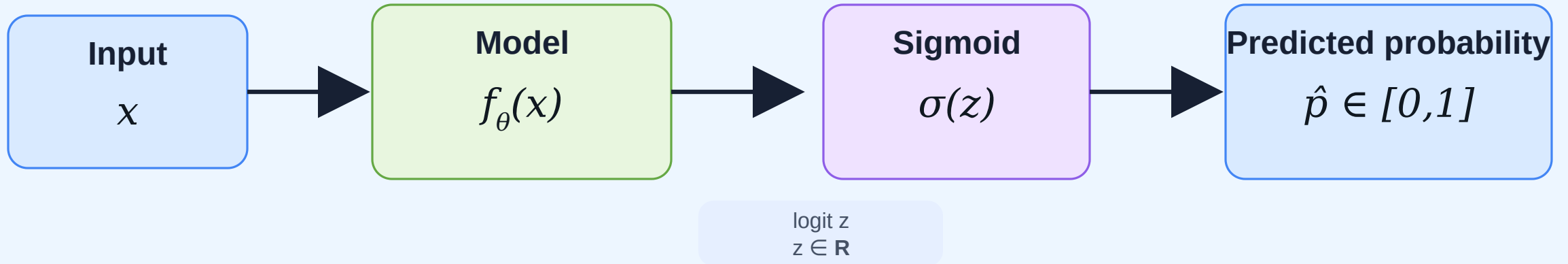
How parameters determine the loss

The loss depends on θ indirectly through the model prediction.



the gradient communicates how the loss changes with respect to θ

Binary classification: logits, sigmoid and BCE



Binary cross-entropy

$$z = f_{\theta}(x)$$

$$\hat{p} = \sigma(z) = 1 / (1 + e^{-z})$$

$$L = -[y \log(\hat{p}) + (1-y) \log(1-\hat{p})]$$

$y \in \{0,1\}$

PyTorch

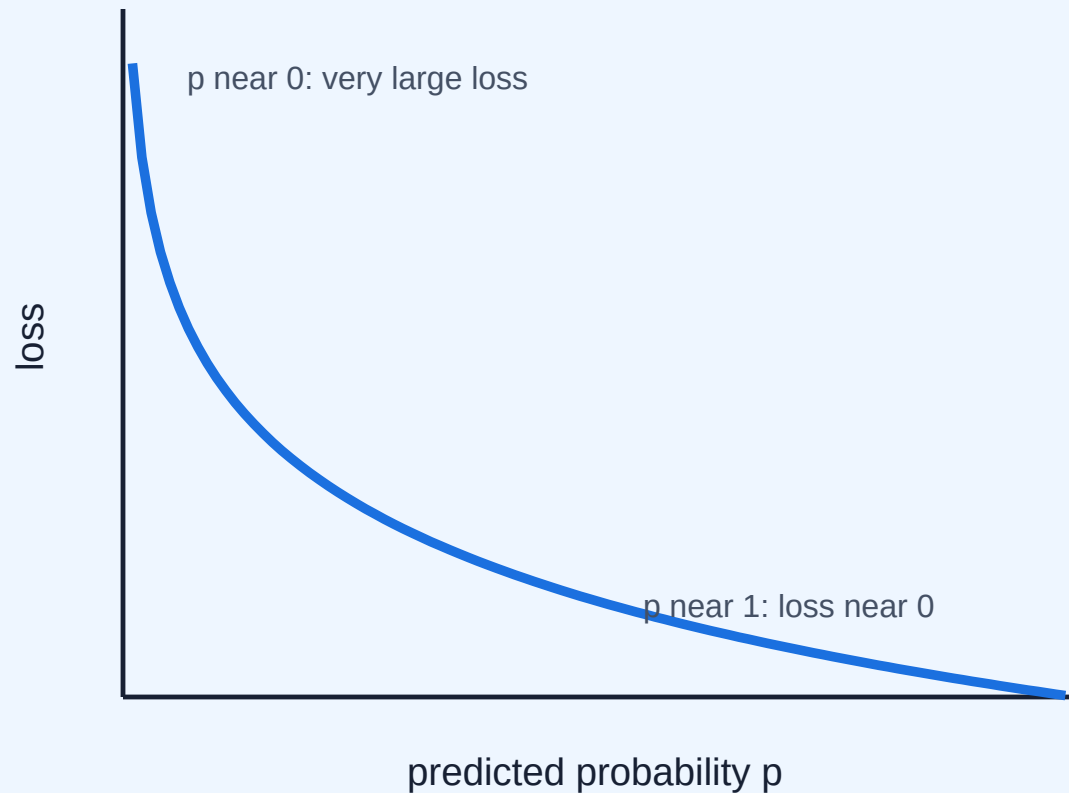
`nn.BCEWithLogitsLoss`

expects logits z ; sigmoid is applied internally.

Binary cross-entropy: the typical loss curves

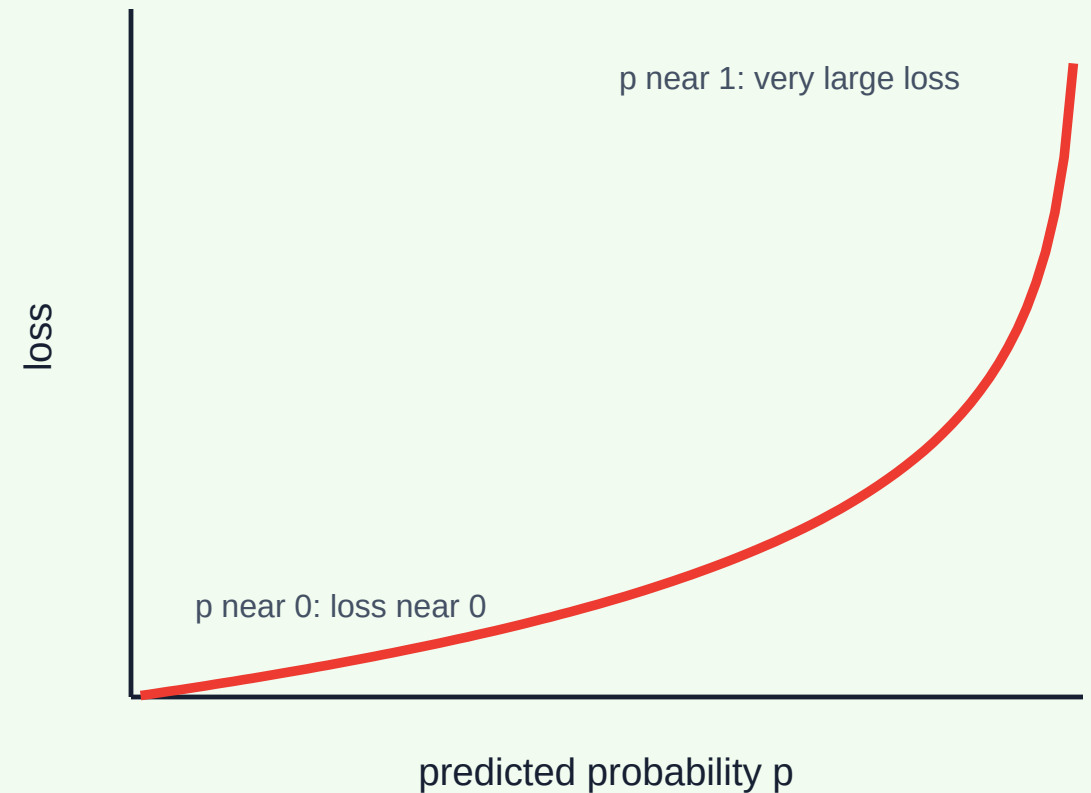
True label $y = 1$

$$L(p, 1) = -\log(p)$$



True label $y = 0$

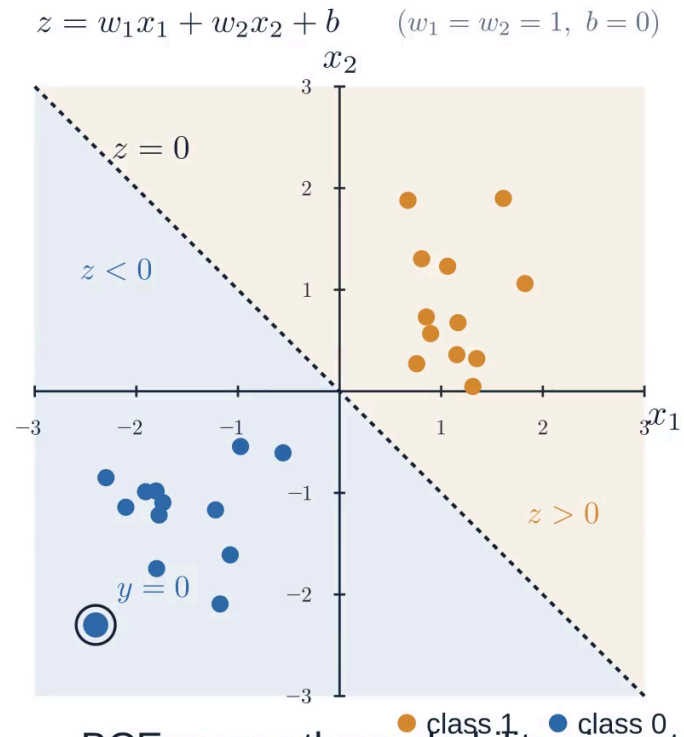
$$L(p, 0) = -\log(1-p)$$



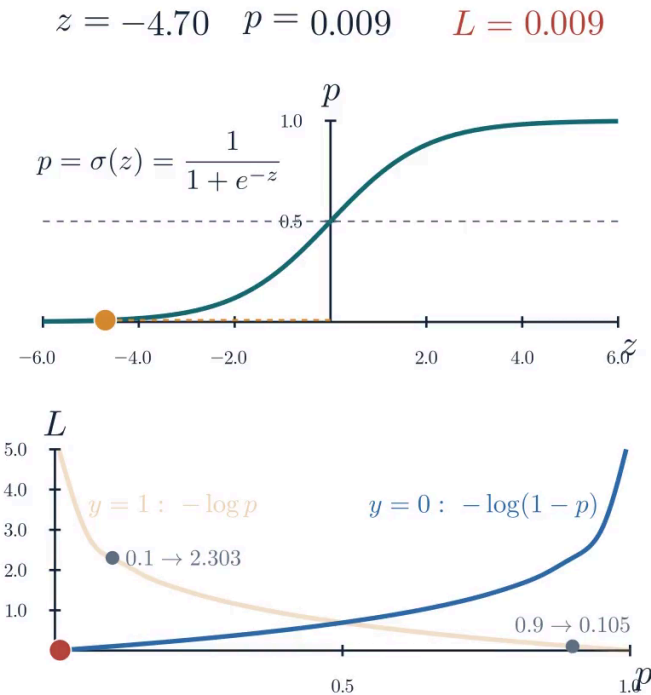
Confident wrong predictions receive a very large penalty.

From score to probability to loss

From score z to probability to loss

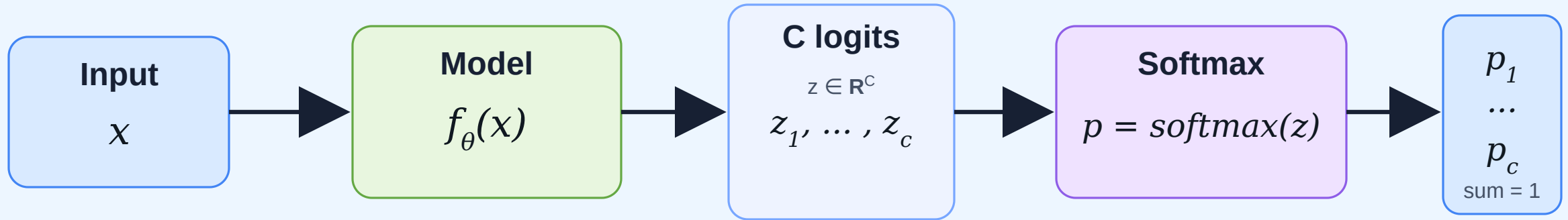


BCE scores the probability given to the true class: p for $y = 1$, $1 - p$ for $y = 0$.



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Multiclass classification: logits, softmax and cross-entropy



Cross-entropy for true class y

$$p_c = e^{z_c} / \sum_k e^{z_k}$$

$$L(z, y) = -\log p_y$$

Raise the true-class logit relative to the others.

PyTorch

`nn.CrossEntropyLoss`

expects raw logits (B,C) and class indices 0...(C-1).

Cross-entropy intuition: confidence matters

Binary classification

True label $y = 1$

Prediction	Interpretation	Loss
$p = 0.9$	correct, high confidence	0.105
$p = 0.6$	correct, low confidence	0.511
$p = 0.1$	wrong, high confidence	2.303

Multiclass classification

True class $y = 2$

Predicted probabilities	Interpretation	Loss
$[0.1, 0.8, 0.1]$	correct high confidence	0.223
$[0.2, 0.5, 0.3]$	correct low confidence	0.693
$[0.7, 0.2, 0.1]$	wrong high confidence	1.609

More probability on the true class means lower cross-entropy.

What the loss function expects in PyTorch

Task	Model output	Target	Loss in PyTorch	Shapes
Regression	$\hat{y} \in \mathbf{R}$	$y \in \mathbf{R}$	<code>nn.MSELoss</code>	output: (B,) or (B,1) target: same shape dtype: float
Binary classification	$z \in \mathbf{R}$ $\hat{p} = \sigma(z)$	$y \in \{0,1\}$	<code>nn.BCEWithLogitsLoss</code> expects logits, not probabilities	logits: (B,) or (B,1) target: same shape dtype: float
Multiclass classification	$z \in \mathbf{R}^C$ $p = \text{softmax}(z)$	$y \in \{0, \dots, C-1\}$	<code>nn.CrossEntropyLoss</code> expects logits and class indices	logits: (B,C) target: (B,) dtype: long

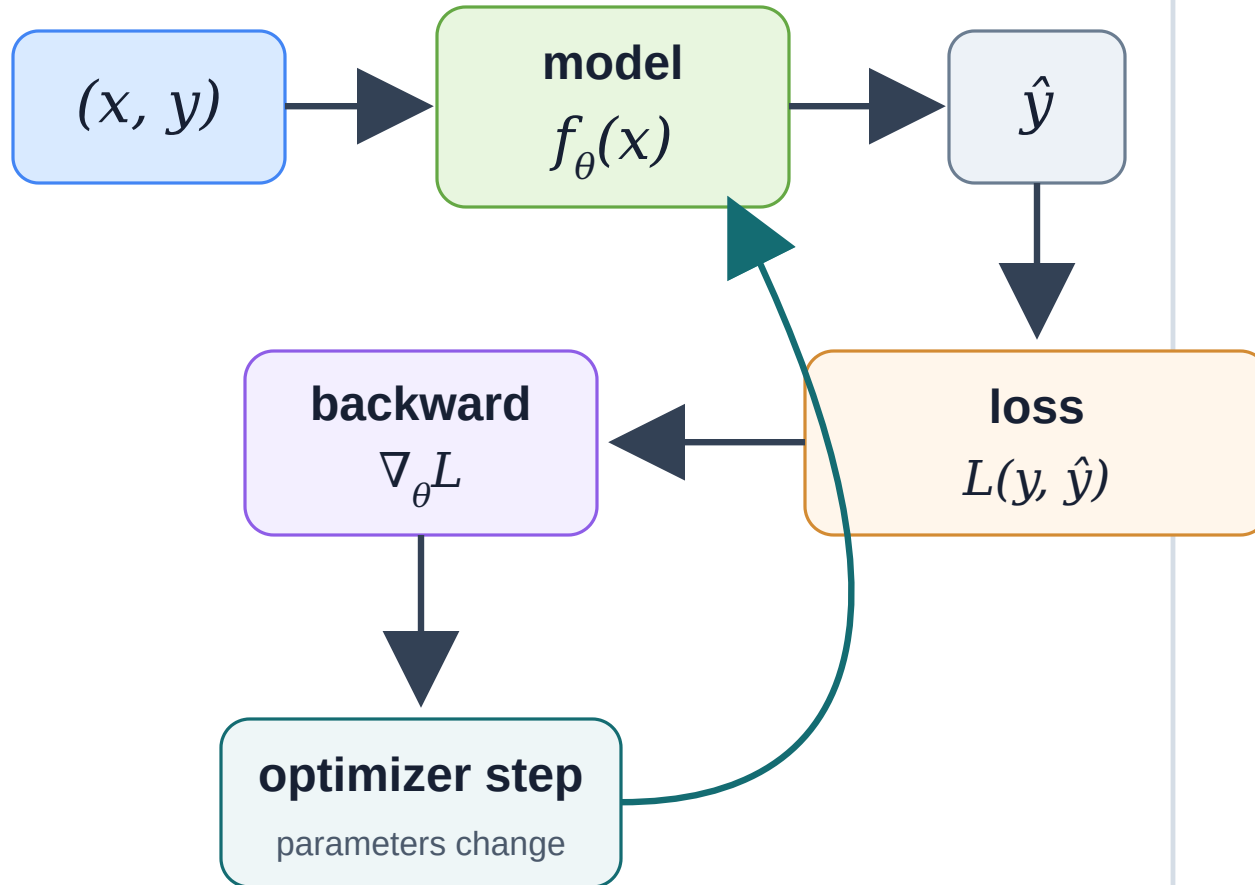
Use sigmoid/softmax to interpret predictions; train with raw logits.

Training is not inference

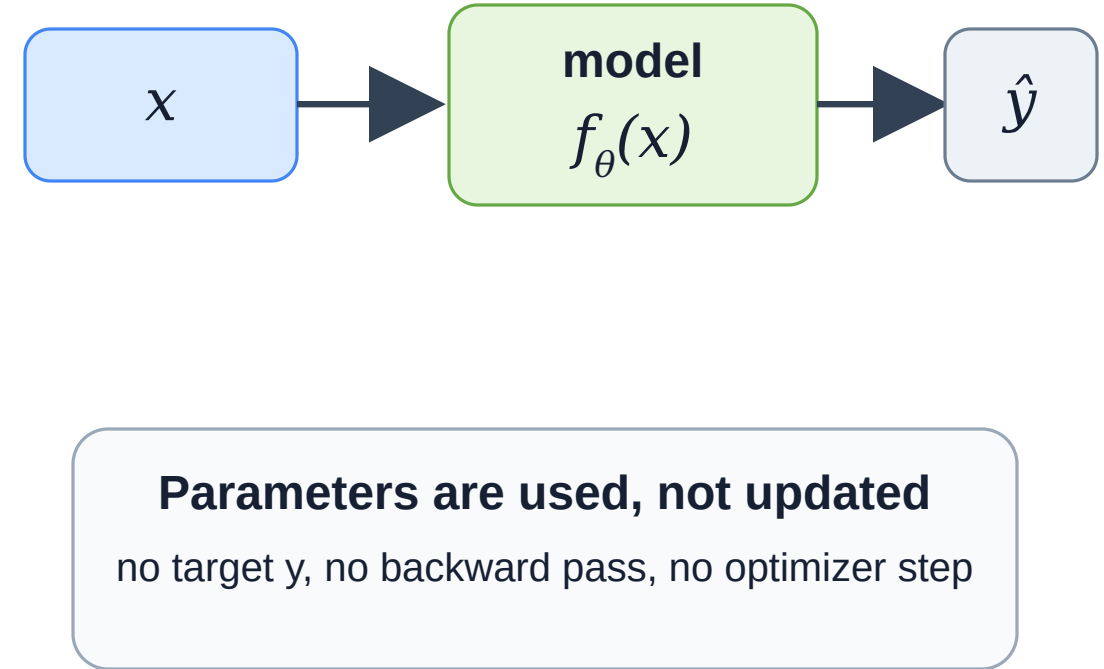
Inference	Training
$x \rightarrow f_{\theta}(x) \rightarrow \hat{y}$	$(x, y) \rightarrow f_{\theta}(x) \rightarrow \hat{y} \rightarrow L \rightarrow \theta'$
parameters are used	parameters are changed

Training versus inference

Training



Inference



One parameter

Suppose:

$$\hat{y} = wx$$

Changing w changes the prediction and therefore changes the loss.

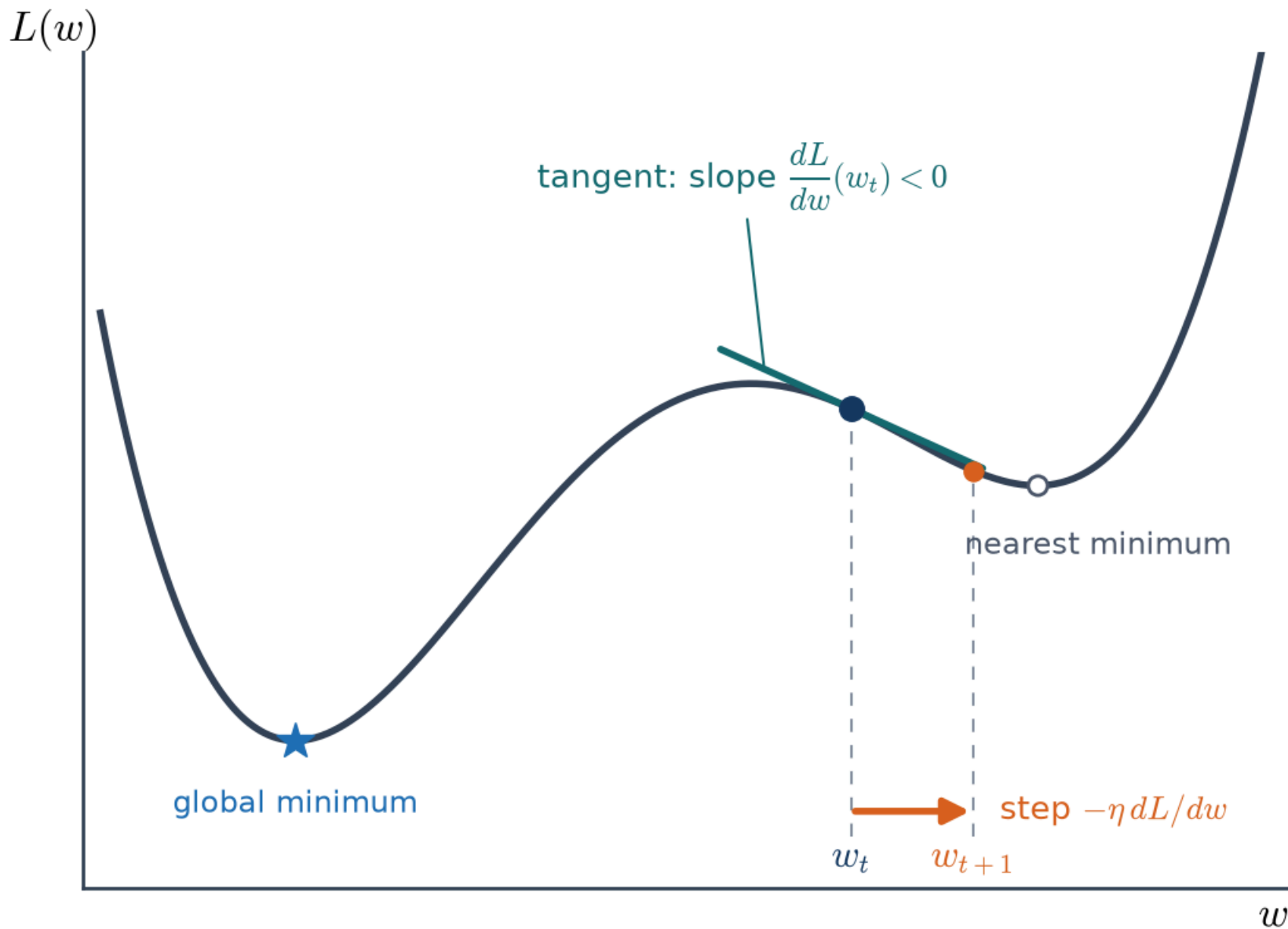
The gradient

$$\frac{\partial L}{\partial w}$$

The gradient gives **local information** about how the loss changes.

The derivative is local information

At the current value w_t , the slope says how L changes nearby, and nothing more.



What the gradient tells us

- the direction in which L decreases at w_t
- how steep it is there

What it does not tell us

- where the minimum is
- whether a better minimum exists elsewhere

Gradient descent

$$w_{t+1} = w_t - \eta \frac{\partial L}{\partial w}$$

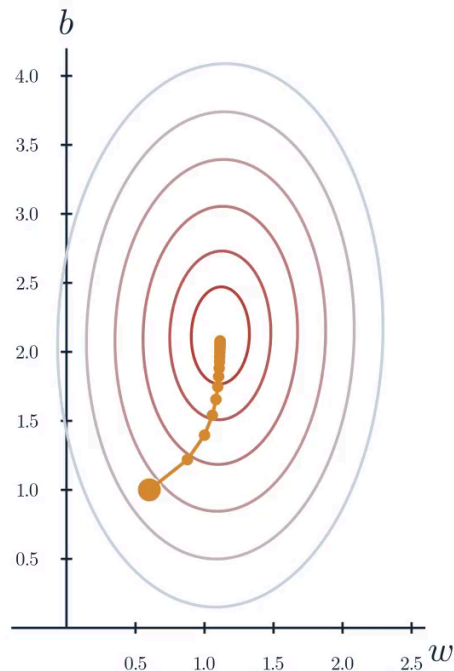
Two ingredients:

- **direction** from the gradient
- **step size** from the learning rate

Derivative, gradient and update

Derivative, gradient and update

MSE of the regression above: level curves of $J(w, b)$



$$\nabla J(w, b) = \begin{bmatrix} \partial J / \partial w \\ \partial J / \partial b \end{bmatrix}$$

$$\nabla J(0.6, 1.0) \approx \begin{bmatrix} -2.76 \\ -2.19 \end{bmatrix}$$

Arrows have fixed length: they show direction only.

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} J(\theta_t), \quad \eta = 0.1$$

$$t = 15 \quad J(\theta_t) = 0.13$$

Each step uses the gradient at the current point. The path bends toward the minimum.

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Learning rate

η

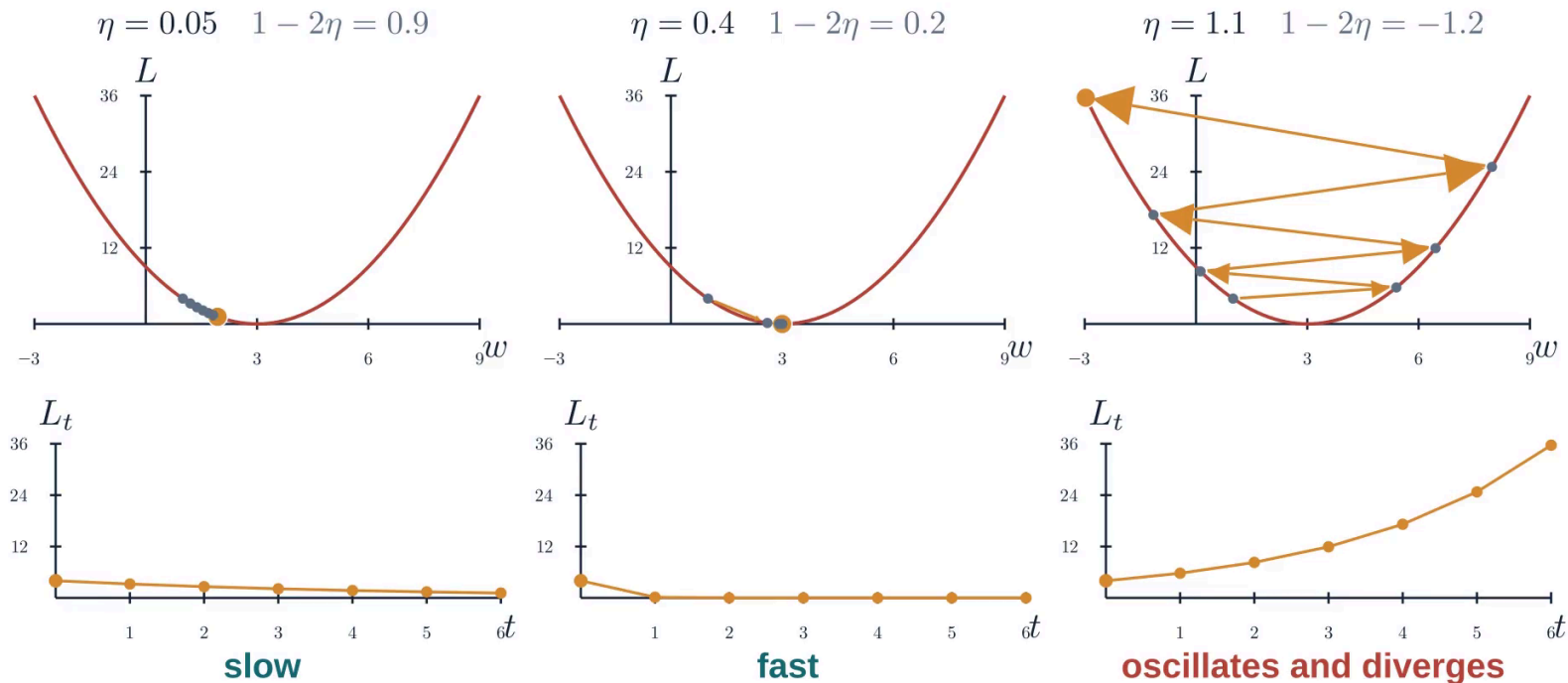
What changes if the step is:

- too small?
- appropriate?
- too large?

Comparing learning rates

Learning rate

$$w_{t+1} - 3 = (1 - 2\eta)(w_t - 3), \quad w_0 = 1 \quad t = 6$$



Here it converges if $0 < \eta < 1$. These values belong to this function and scale; they are not a recipe for networks.

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More than one parameter

$$\theta = (w_1, w_2, \dots, w_p)$$

$$\nabla_{\theta} L = \begin{bmatrix} \partial L / \partial w_1 \\ \vdots \\ \partial L / \partial w_p \end{bmatrix}$$

From example loss to dataset objective

For one observation:

$$L_i(\theta) = L(y_i, f_\theta(x_i))$$

For the full dataset:

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N L_i(\theta)$$

Convention: L is a per-example loss; J is the dataset objective.

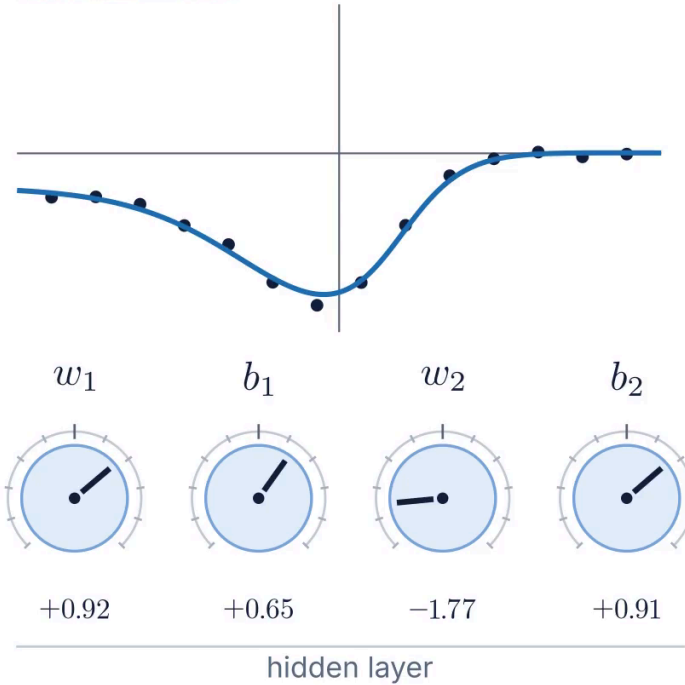
Each parameter is a knob

Each parameter is a knob

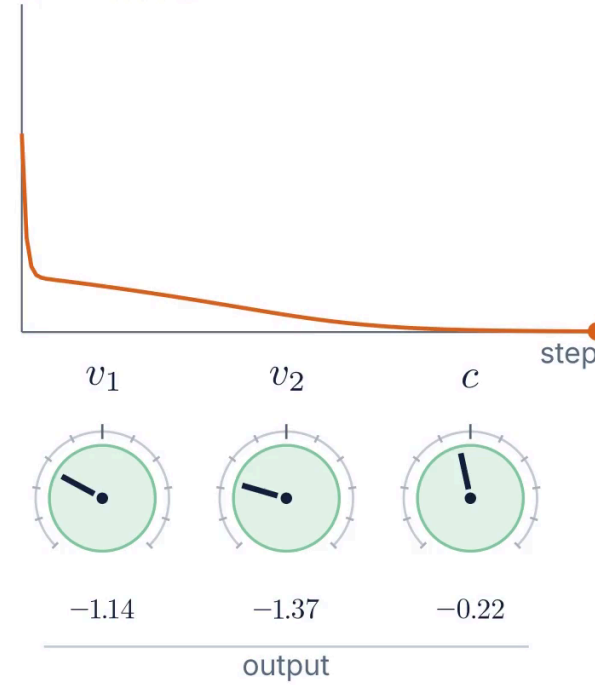
$$f(x) = v_1 \tanh(w_1 x + b_1) + v_2 \tanh(w_2 x + b_2) + c$$

At the minimum $\nabla_{\theta} J \approx 0$: there is nowhere left to turn.

data and model



$J = 0.004$



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Batch, SGD and mini-batch

For a subset $\mathcal{B}$ of the data:

$$J_{\mathcal{B}}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} L_i(\theta)$$

- **batch:** $\mathcal{B} = \mathcal{D}$
- **SGD:** $|\mathcal{B}| = 1$
- **mini-batch:** $1 < |\mathcal{B}| < N$

The update uses $g_t = \nabla_{\theta} J_{\mathcal{B}_t}(\theta_t)$.

$$J_B(\theta) = \frac{1}{|B|} \sum_{i \in B} \mathcal{L}_i(\theta)$$

B = subset used to estimate the gradient at step t

Batch gradient descent

Dataset $\mathcal{D}$ (N examples)



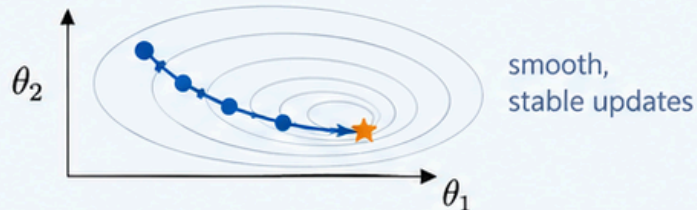
$$B = \mathcal{D}$$

(use all N examples)

$$g_t = \nabla_{\theta} J_{\mathcal{D}}(\theta)$$

uses all examples
before each update

- low gradient noise
- high computation per step



Stochastic gradient descent (SGD)

Dataset $\mathcal{D}$ (N examples)



selected example i at step t

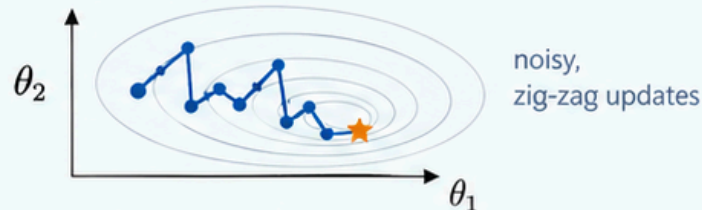
$$|B| = 1$$

(one example)

$$g_t = \nabla_{\theta} \mathcal{L}_i(\theta)$$

uses one example
per update

- high gradient noise
- cheap updates



Mini-batch gradient descent

Dataset $\mathcal{D}$ (N examples)



mini-batch B_t at step t

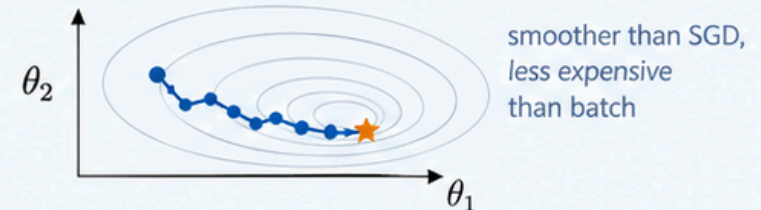
$$1 < |B| < N$$

(small subset)

$$g_t = \nabla_{\theta} J_{B_t}(\theta)$$

uses a small subset
per update

- balanced noise / cost
- GPU-friendly



Batch = accurate but expensive • **SGD = noisy but fast** • **Mini-batch = practical default in deep learning**

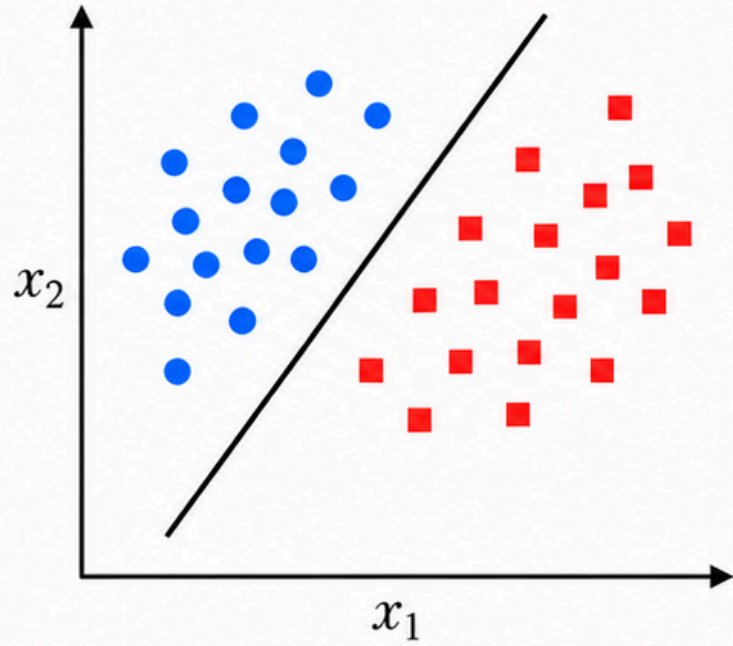
Optimization is a trajectory

$$\theta_{t+1} = \theta_t - \eta g_t$$
$$\theta_0, \theta_1, \theta_2, \dots$$

Training moves through parameter space using successive gradient estimates.

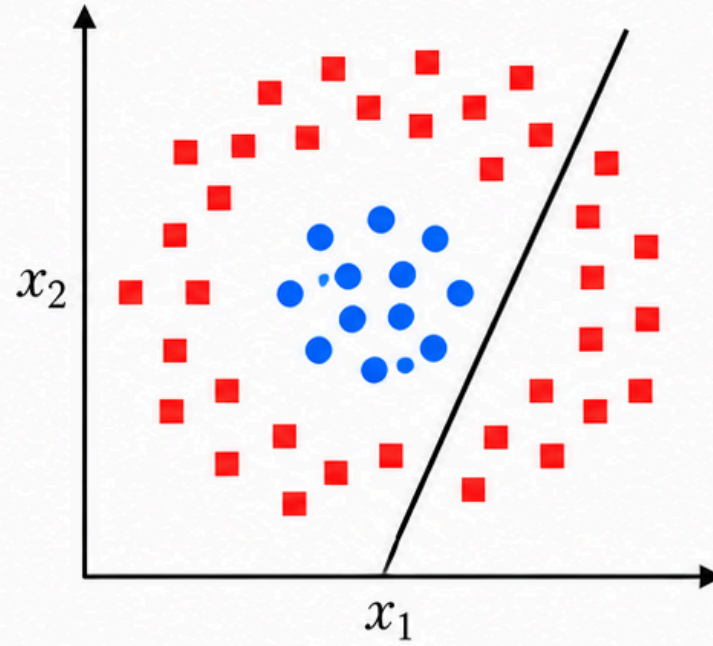
Linearly separable data

A linear model is sufficient.



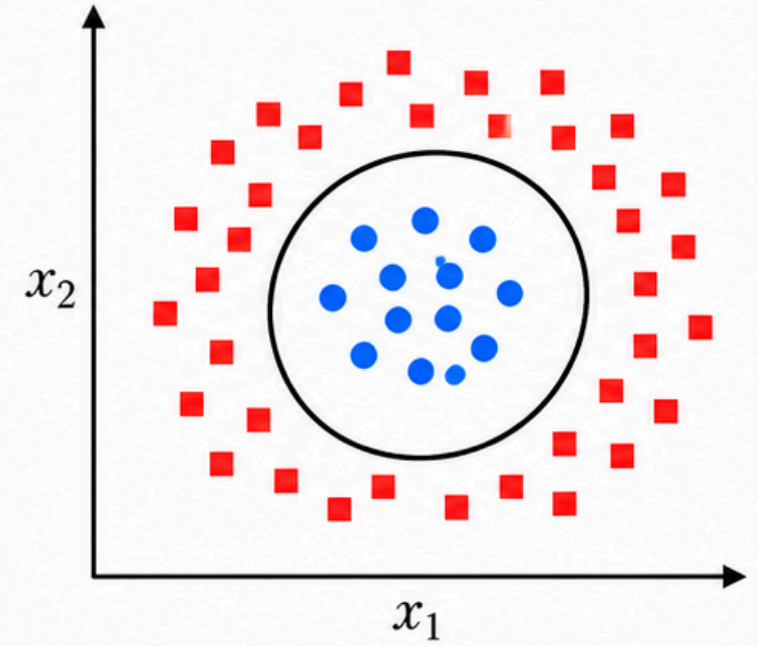
Non-linearly separable data

A linear model cannot represent the boundary.



Non-linear model

Hidden layers + nonlinearity can fit the structure.



Optimization can only fit functions that the model class can represent.

Stacking linear transformations

$$h = W_1x + b_1$$

$$\hat{y} = W_2h + b_2$$

Therefore:

$$\hat{y} = W_2W_1x + W_2b_1 + b_2$$

Several linear layers can collapse into a single affine transformation.

Nonlinearity changes the game

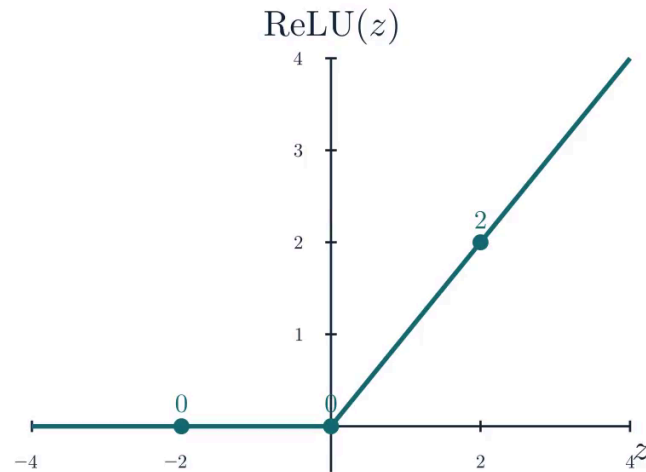
$$z = Wx + b, \quad a = \phi(z)$$

A nonlinear activation prevents the composition from collapsing into one linear map.

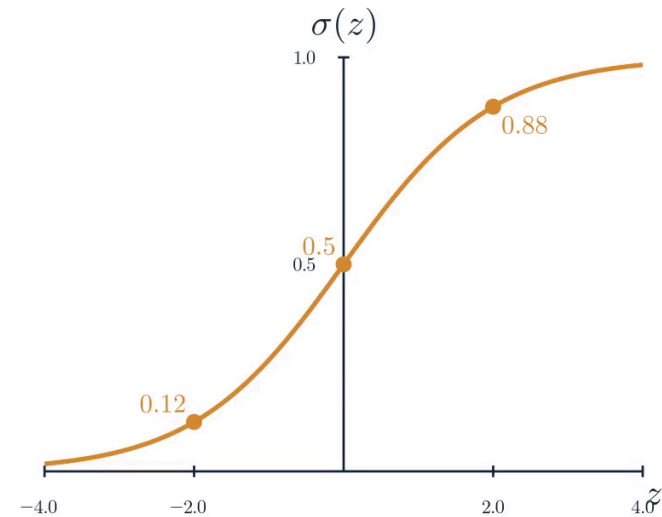
Why we need an activation

Why we need an activation

$\text{ReLU}(z) = \max(0, z)$
range $[0, \infty)$ · hidden layers



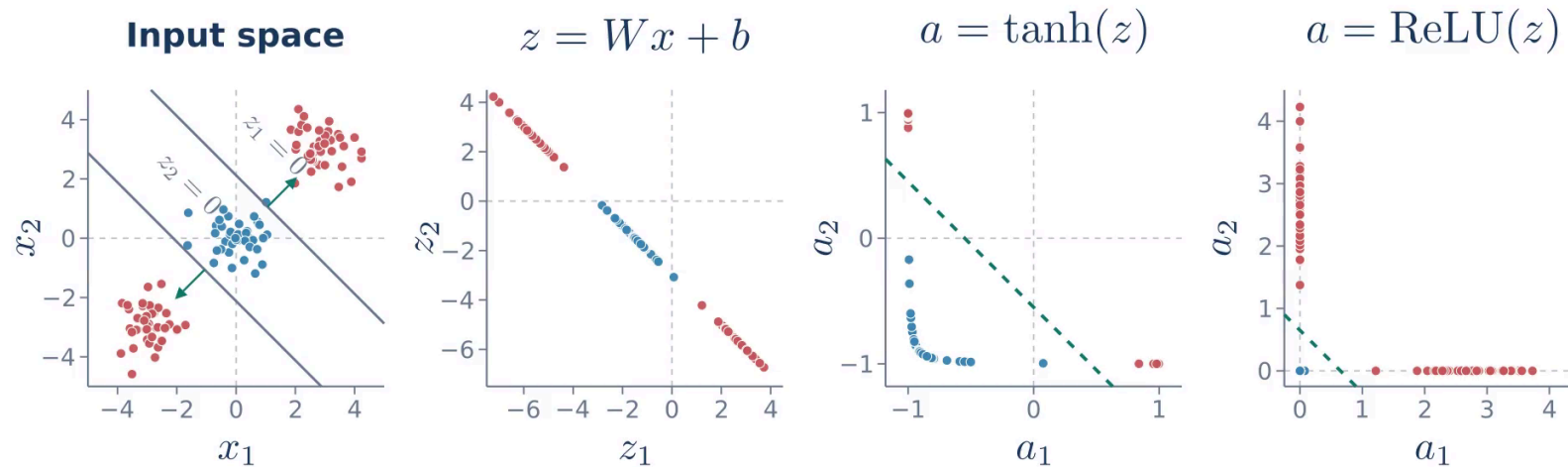
$\sigma(z) = \frac{1}{1 + e^{-z}}$
range $(0, 1)$ · output probability



ReLU: 0, 0, 2 (no upper bound; not differentiable at 0). Sigmoid: 0.12, 0.5, 0.88.

[Open animation](#)

Affine maps and nonlinear activations



$$z_1 = (x_1 + x_2)/\sqrt{2} - 1.5, \quad z_2 = -(x_1 + x_2)/\sqrt{2} - 1.5$$

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ReLU

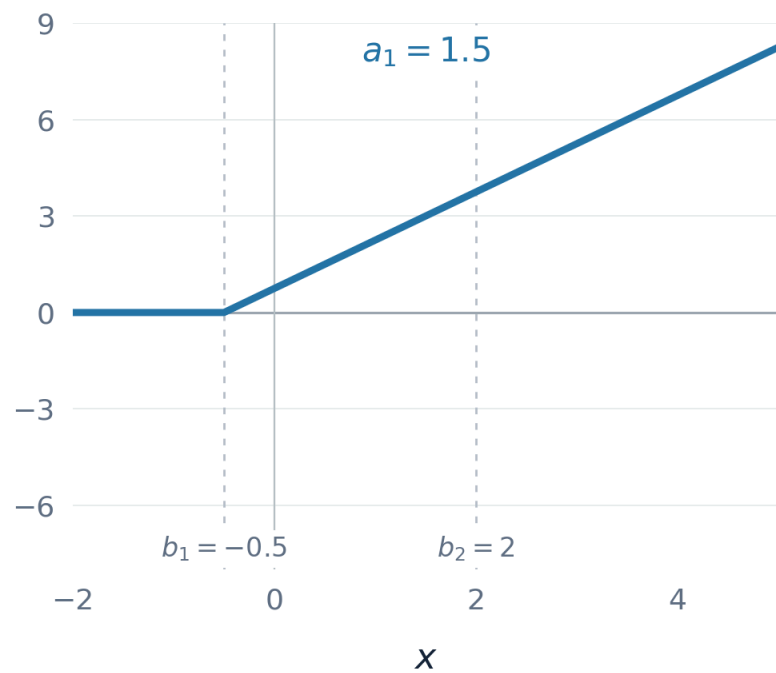
$$\text{ReLU}(z) = \max(0, z)$$

A very simple nonlinear function can become a powerful building block.

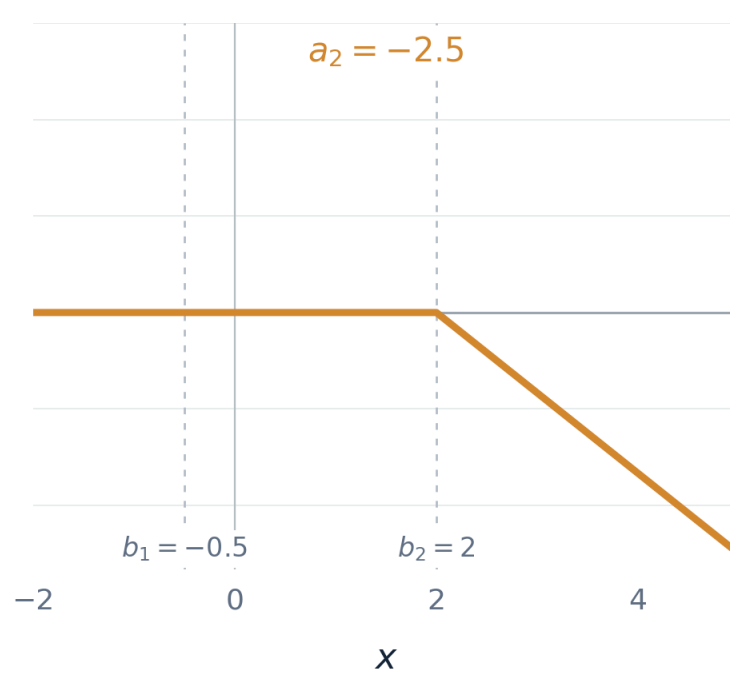
One ReLU as a building block

A single unit produces a simple piecewise-linear transformation.

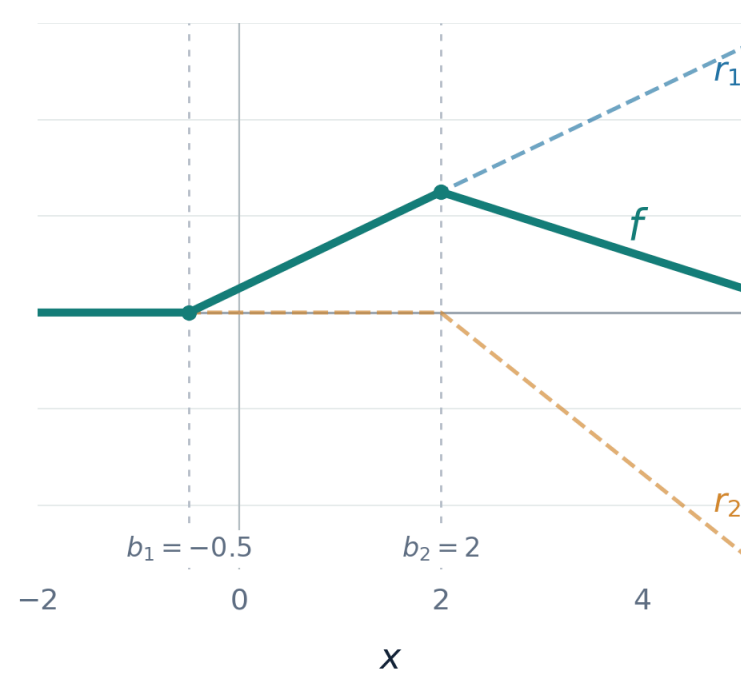
$$r_1(x) = a_1 \text{ReLU}(x - b_1)$$



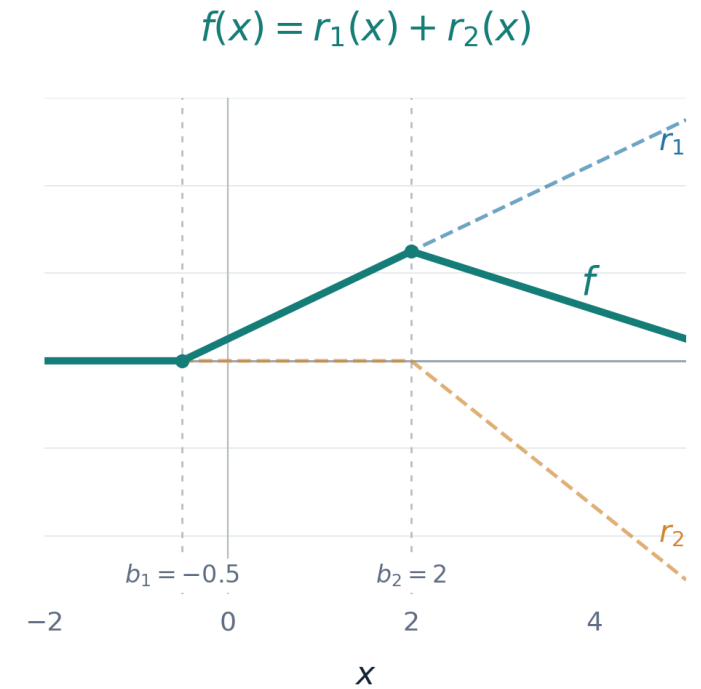
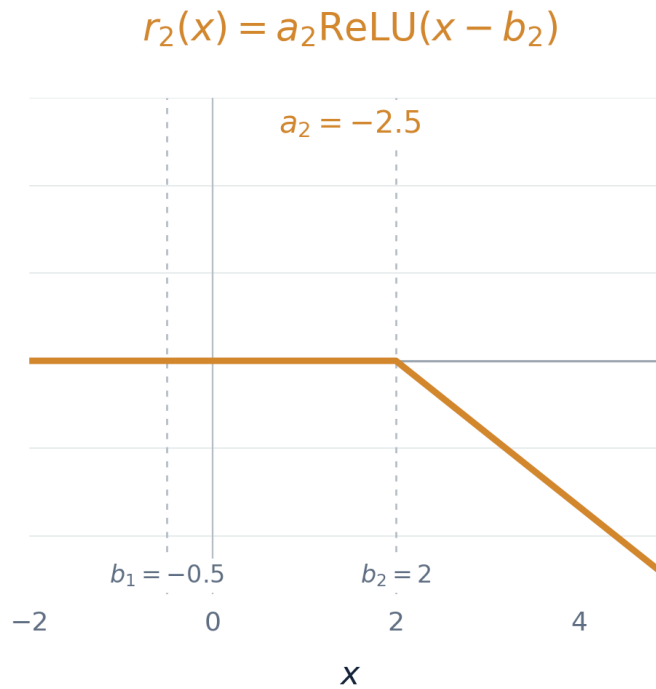
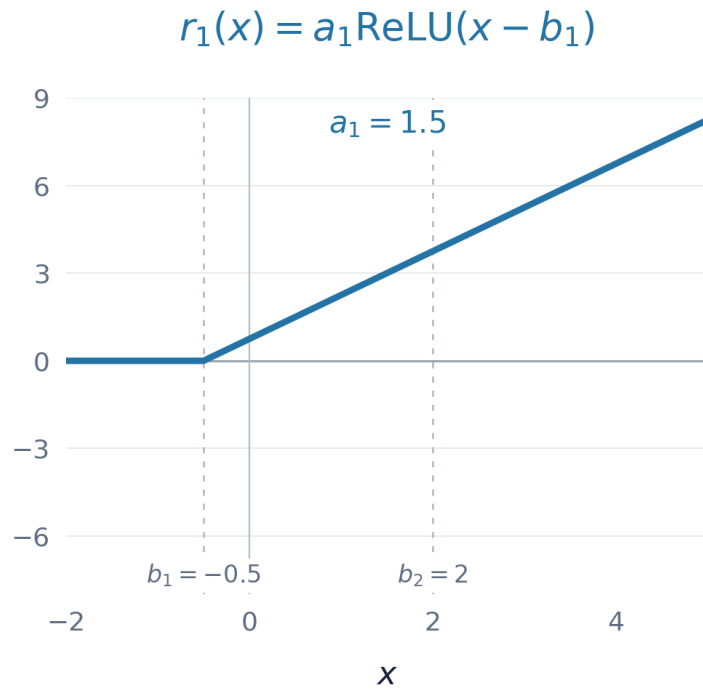
$$r_2(x) = a_2 \text{ReLU}(x - b_2)$$



$$f(x) = r_1(x) + r_2(x)$$



Approximation with ReLU components



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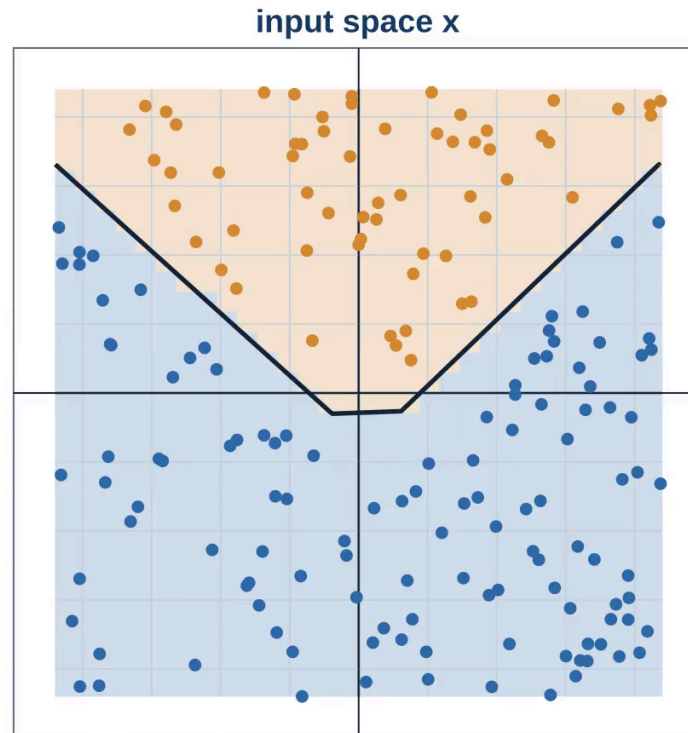
A hidden layer

$$z_1 = W_1x + b_1, \quad a_1 = \phi(z_1)$$
$$\hat{y} = W_2a_1 + b_2$$

A hidden layer learns an **intermediate representation**.

Hidden representation and decision boundary

Hidden representation and decision boundary



1. $x \in \mathbb{R}^2$
2. $z_1 = W_1 x + b_1$
3. $a_1 = \text{ReLU}(z_1)$
4. $w_2^\top a_1 + b_2 = 0$
5. boundary in x

Trained 2–2–1 MLP, not hand-picked weights
numpy · seed 3 · 6000 GD steps · $\eta = 0.1$
training accuracy: 100 %

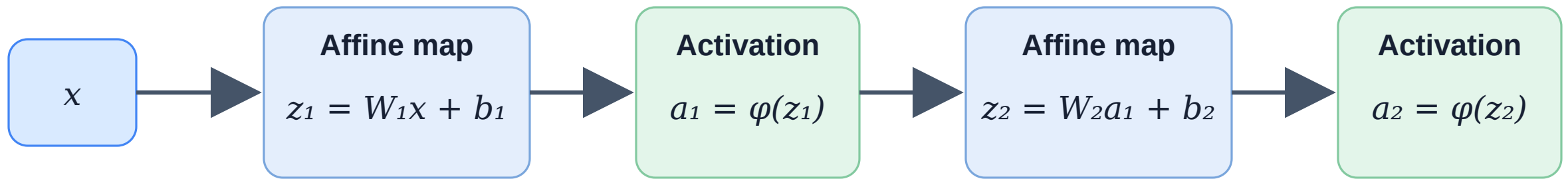
● class 1 ● class 0

With seeds 2 and 7 the same training stops at 91–93 %: capacity does not guarantee good training.

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MLP as a composition of affine maps and activations

φ denotes a generic nonlinear activation; σ is reserved for sigmoid.

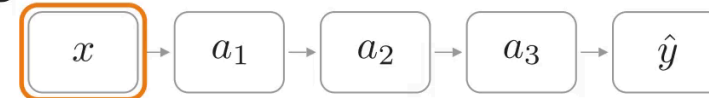
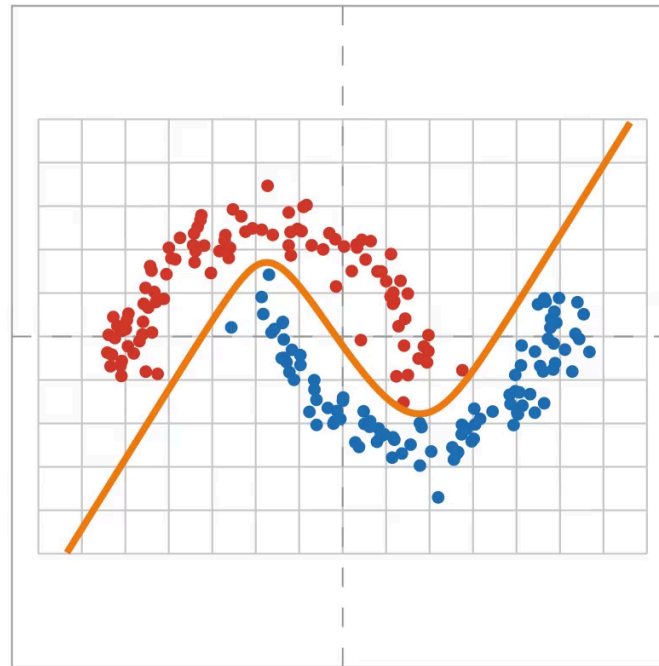


Without φ , stacked affine maps collapse into one affine map.

Each layer transforms the space

Each layer transforms the space

Trained 2-2-2-2-1 MLP · training accuracy 100%



$$\hat{y} = \sigma(w_4^\top a_3 + b_4)$$

the output layer is linear:
one straight line in a_3

The same line, seen in x ,
is a curved boundary.

Hidden layers bend the space; the output layer only draws a straight line.

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Forward pass

$$x \rightarrow z_1 \rightarrow a_1 \rightarrow z_2 \rightarrow a_2 \rightarrow \hat{y} \rightarrow L$$

The forward pass evaluates the composed function.

Computational graph

Each hidden layer contains an affine map and an activation:

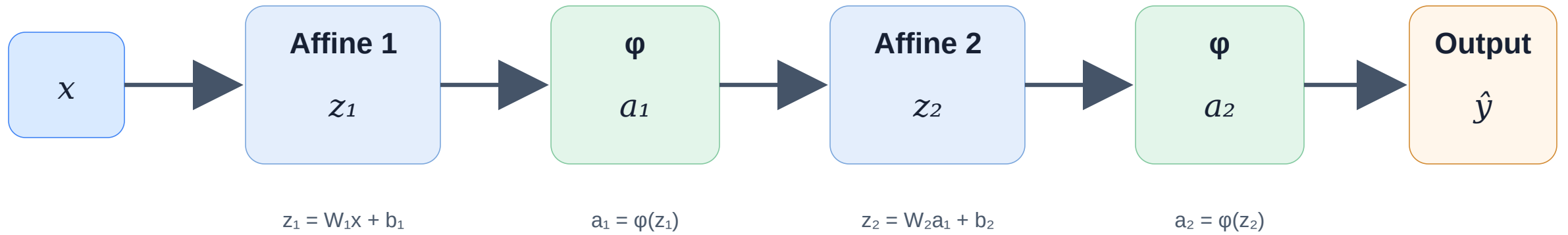
$$z_l = W_l a_{l-1} + b_l, \quad a_l = \phi(z_l), \quad a_0 = x$$

For a linear output and squared loss:

$$\hat{y} = W_3 a_2 + b_3, \quad L = \frac{1}{2} \|\hat{y} - y\|^2$$

Forward pass through affine and nonlinear layers

Forward computes values left to right and stores intermediate values for backward.



Saved for backward

x, z_1, a_1, z_2, a_2 and $\hat{y}$

Chain rule

If:

$$L = L(\hat{y}(a(w)))$$

then:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial a} \frac{\partial a}{\partial w}$$

Backward pass

The same graph is traversed in the opposite direction to propagate gradients.

Backward pass: signals and parameter gradients

For a linear output and squared loss: $L = 1/2 ||\hat{y} - y||^2$

Backward starts at the loss

$$\delta_{\hat{y}} = \partial L / \partial \hat{y} = \hat{y} - y$$

seed gradient at the model output

Through activation ϕ

$$\delta_{z_1} = \delta_{a_1} \odot \phi'(z_1)$$

Through hidden affine

$$\delta_{a_1} = W_2^T \delta_{z_2}$$

Through activation ϕ

$$\delta_{z_2} = \delta_{a_2} \odot \phi'(z_2)$$

Through output layer

$$\delta_{a_2} = W_3^T \delta_{\hat{y}}$$

Parameter gradients from the same backward signals

Output layer

$$\nabla_{W_3} L = \delta_{\hat{y}} a_2^T$$

$$\nabla_{b_3} L = \delta_{\hat{y}}$$

Hidden layer 2

$$\nabla_{W_2} L = \delta_{z_2} a_1^T$$

$$\nabla_{b_2} L = \delta_{z_2}$$

Hidden layer 1

$$\nabla_{W_1} L = \delta_{z_1} x^T$$

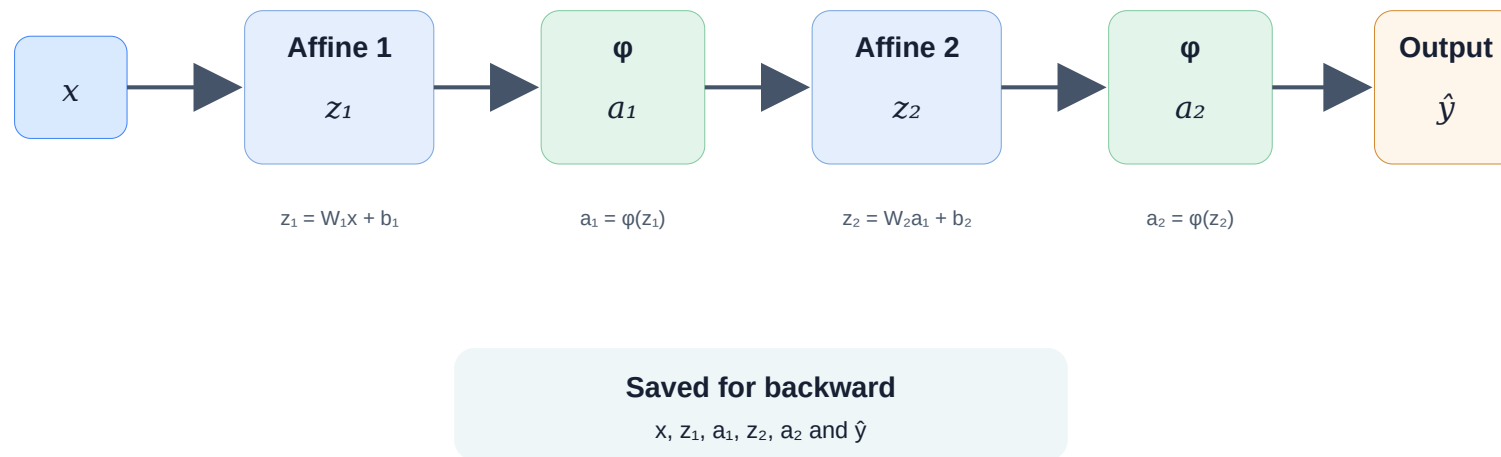
$$\nabla_{b_1} L = \delta_{z_1}$$

Backward propagates signals; each affine layer uses its signal to compute gradients for both W and b.

Forward values and backward signals

Forward pass through affine and nonlinear layers

Forward computes values left to right and stores intermediate values for backward.



[Open animation](#)

Gradients for every parameter

Each trainable parameter receives:

$$\frac{\partial L}{\partial w_i}$$

The backward pass produces the information required by the optimizer.

Backpropagation is not gradient descent

Backpropagation	Optimizer
computes $\nabla_{\theta} J_{\mathcal{B}}$	uses $\nabla_{\theta} J_{\mathcal{B}}$
differentiates the graph	updates the parameters

backward $\neq$ update

Automatic differentiation

Manual view:

operations $\rightarrow$ local derivatives $\rightarrow$ chain rule

PyTorch:

```
loss.backward()
```

Autograd automates differentiation of the computational graph.

The PyTorch training loop

```
optimizer.zero_grad()

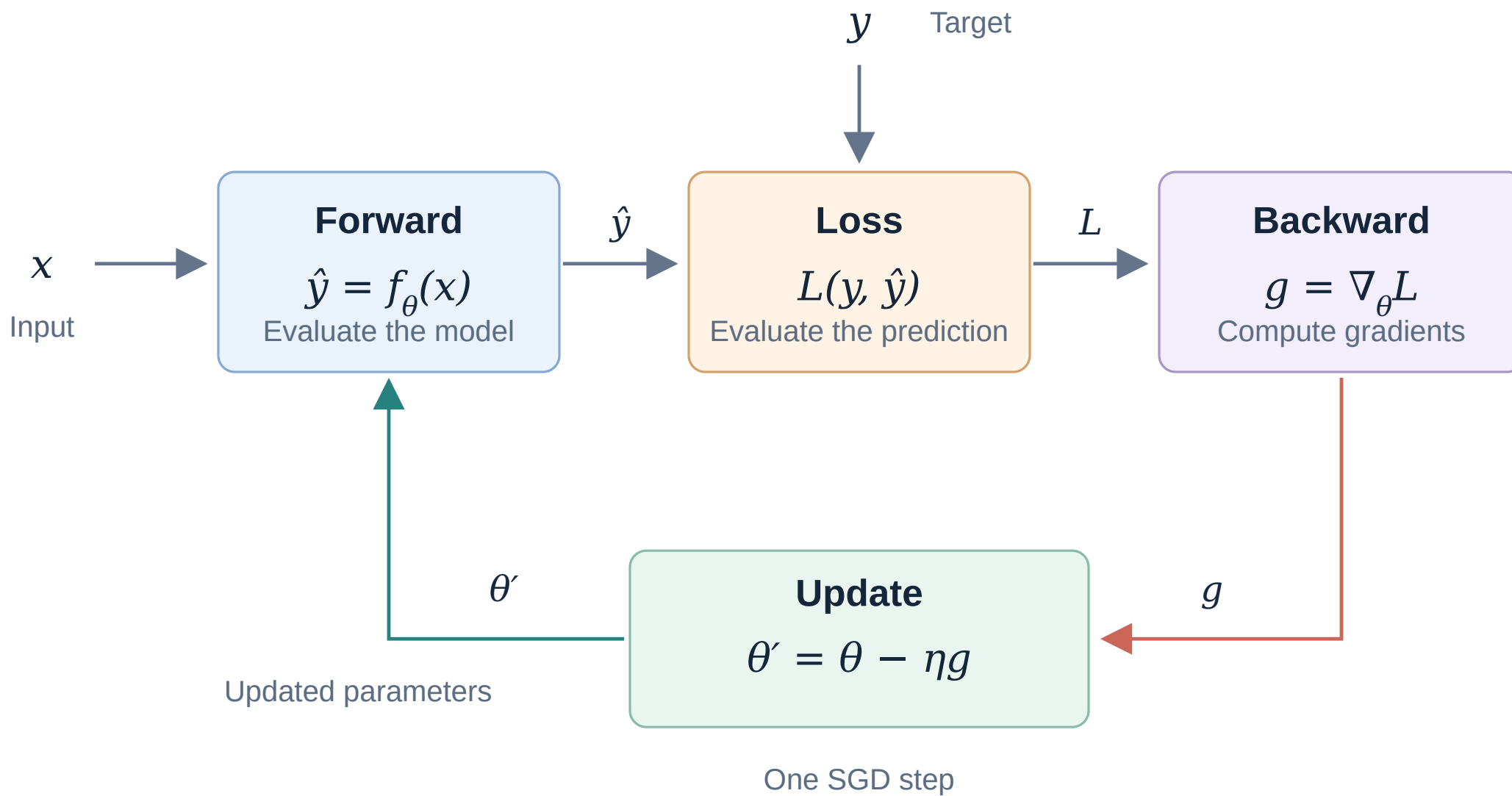
y_hat = model(x)
loss = criterion(y_hat, y)

loss.backward()
optimizer.step()
```

From mathematics to PyTorch

Each training-loop operation has a direct implementation.

Concept	Mathematics	PyTorch
forward	$\hat{y} = f_{\theta}(x)$	<code>y_hat = model(x)</code>
loss	$J_B(\theta) = \text{mean}_{i \in B} L(y_i, \hat{y}_i)$	<code>loss = criterion(...)</code>
gradient	$\nabla_{\theta} J_B$	<code>loss.backward()</code>
update	$\theta' = \theta - \eta \nabla_{\theta} J_B$	<code>optimizer.step()</code>



What changes in deep learning?

The learning loop remains the same.

What changes is:

- f_{θ}
- architecture
- loss
- optimizer
- data
- scale

CNNs, RNNs, ViTs and Transformers do **not** replace the learning loop.

Exit question

A student says:

“Backpropagation updates the weights to minimize the loss.”

What is incorrect in that statement?

Guided practice — PyTorch in one hour

[Open notebook](#)

- tensors and affine transformations
- scalar autograd and parameter updates
- `TensorDataset` and `DataLoader`
- MLP for nonlinear classification
- loss curves and decision boundaries

Predict the behavior before running each experiment.

Exercise — Multiclass MLP with tabular data

[Open notebook](#)

Dry Bean dataset from Kaggle: 16 attributes and 7 classes.

- split and standardize the data
- create a `DataLoader`
- implement an MLP and the training loop
- use `CrossEntropyLoss`
- evaluate accuracy, macro F1 and the confusion matrix

Session 01 — Mental model

forward $\rightarrow$ loss $\rightarrow$ backward $\rightarrow$ update

If this loop is clear, the rest of the course has a common foundation.